

### Question no-3(PHP+Mysql)/ q3.php:

```
<?php
$conn = new mysqli("127.0.0.1", "root", "",
"database_name", 3306);
$query1=" paste here";

$store = $conn->query($query1);

while ($show = $store->fetch_assoc()) {
    echo $show['CategoryName'] . " --- " .
    $show['total_rev'] . "<br>";
}
?>
```

#### Q1 — always

COUNT + GROUP BY (+ HAVING)

```
SELECT Status,COUNT(*) AS total
FROM book_loans
GROUP BY Status
HAVING COUNT(*) > 1;
```

No HAVING needed if question says "all statuses"

#### Q2 — always

UPDATE WITH WHERE CONDITION

```
UPDATE book_loans
SET Status = 'Grace
Period',PenaltyFee=0
WHERE Status = 'Overdue'
AND DaysOverdue < 7;
```

Swap column names + values from the question

#### Q3 — always

CONDITIONAL UPDATE WITH CAP (BONUS)

```
UPDATE book_loans
SET PenaltyFee = PenaltyFee * 1.10
WHERE PenaltyFee > 20
AND (PenaltyFee * 1.10) <= 50
```

Cap check goes inside WHERE — not in PHP

#### Q4 — always

SELECT WITH LABEL + ORDER BY

/\* simple: SUM + sort \*/

```
SELECT BookTitle,
SUM(PenaltyFee) AS total
FROM book_loans
GROUP BY BookTitle
ORDER BY total DESC;
```

#### Useful clauses:

```
GROUP BY col
HAVING COUNT(*) > 1
ORDER BY col DESC
WHERE col > val
AND / OR
```

#### Aggregate functions:

```
COUNT(*)
SUM(col)
AVG(col)
MAX(col)
MIN(col)
```

#### Like Pattern:

4 PATTERNS TO MEMORISE

```
LIKE 'A%' -- starts with A
LIKE '%n' -- ends with n
```

```
LIKE '%ar%' -- contains ar
LIKE '_a%' -- 2nd letter is a
Example:
SELECT * FROM students
WHERE StudentName LIKE '%ar%';
```

### Question no-2(PHP+HTML)/ q2.php:

```
<body>
    <form action="" method="POST">
        input1: <input type="number"
name="input1"><br><br>
        input2: <input type="number"
name="input2"><br><br>
        input3: <input type="number"
name="input3"><br><br>
        <button name="submit">Submit</button>
    </form>
<?php
    if(isset($_POST['submit'])){
        $input1 = (int)$_POST['input1'];
        $input2 = (int)$_POST['input2'];
        $input3 = (int)$_POST['input3'];
        // --- logic here ---
        $totalpizza =
        ceil(($numberofstudents*$sliceperstudent)/$slic
        eperpizza);
        $leftoverslice= ($totalpizza*$sliceperpizza)-
        ($numberofstudents*$sliceperstudent);
        $wastedmoney =
        (1050/$sliceperpizza)*$leftoverslice;
        // --- Output ---
        echo "<table border='1'>";
        echo "<tr>
            <th>Input1</th>
            <th>Input2</th>
            <th>Input3</th>
            <th>Output1</th>
            <th>Output2</th>
            <th>Output3</th>
        </tr>";
        echo "<tr>
            <td>$input1</td>
            <td>$input2</td>
            <td>$input3</td>
            <td>$output1</td>
            <td>$output2</td>
            <td>$output3</td>
        </tr>";
        echo "</table>";
    }
?>
</body>
```

### Question no-1(Javascript+HTML):

#### q1.html(251):

```
<body>
  Enter your password: <input
type="text" name="" id="input1"><br><br>
  <button onclick="pass()" id="btn">Check
password strength</button>
  <p id="show"></p>
  <script src="q1.js"></script>
</body>
```

#### q1.js(251):

```
let point=0;
function pass(){
  let input1=
document.getElementById("input1").value;
  if(input1.length>=6){
    point+=Math.floor(input1.length/2)*10;
    for(let i=0; i<input1.length;i++){
      if(input1[i]>='A' &&
input1[i]<='Z'){
        point+=15;
        break;}}
    for(let i=0; i<input1.length;i++){
      if(input1[i]>='a' &&
input1[i]<='z'){
        point+=15;
        break;}}
    for(let i=0; i<input1.length;i++){
      if(input1[i]>='0' &&
input1[i]<='9'){
        point+=20;
        break;}}
    for(let i=0; i<input1.length;i++){
      if(input1[i]=='!' ||
input1[i]=='@' || input1[i]=='#' ||
input1[i]=='$' ||
input1[i]=='%' || input1[i]=='^' || input1[i]=='&' |
|input1[i]=='*'){
        point+=25;
        break;}}
    if(point>100){
      document.getElementById("show").inn
erText="Perfect Password";
    }else if(point>90){
      document.getElementById("show").inn
erText="Very_strong";
    }else if(point>70){
      document.getElementById("show").inn
erText="Strong";
    }else if(point>50){
```

```
      document.getElementById("show").inn
erText="Medium";
    }else if(point>30){
      document.getElementById("show").inn
erText="Weak";
    }else{
      document.getElementById("show").inn
erText="Very Weak";}}
    else{
      document.getElementById("show").innerTe
xt="very weak";}}
```

#### q1.html(252):

```
<body>
  Enter calory: <input type="number"
id="input1">
  <br><br>
  <button
onclick="pass()",id="pass">Addcalory</button>
  <br><br>
  <p id="total_calory"></p>
  <p id="entry"></p>
  <p id="feedback_message"></p>
  <script src="q1.js"></script>
</body>
```

#### q1.js(252):

```
const calory_goal=2000;
let total_calory =0;
let count =0;
function pass(){
  count++;
  let input1=
Number(document.getElementById('input1').value);
  total_calory+=input1;
  if(total_calory>=0 && total_calory<=800){
    document.getElementById('feedback_message').i
nnerText="You are of to a healthy start";
  }
  else if(total_calory>=801 && total_calory<=1600){
    document.getElementById('feedback_message').i
nnerText="Good progress, keep it balanced!";
  }
  else if(total_calory>=1601 &&
total_calory<=1999){
    document.getElementById('feedback_message').i
nnerText="Almost at your limit!";
  }
  else{
    document.getElementById('feedback_message').i
nnerText="Goal reached!stay mindful!";
  }
  if(count>=10 && total_calory<calory_goal){
    document.getElementById('feedback_message').in
nerText="Be cautious of frequent snacking";
  }
  document.getElementById('total_calory').innerText
="Total calory:"+total_calory;
  document.getElementById('entry').innerText="Entry
: "+count;

  document.getElementById('input1').value="";}
```